Personalized Adaptive Assistance With Reinforcement Learning Control Enhances Engagement, Performance, and Retention in Robot-Assisted Arm-Reaching Exercises

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***Abstract*— This study introduces a new Reinforcement Learning Assist-as-Needed (RL-AAN) controller intended for robot-assisted upper-limb rehabilitation after stroke, which leverages a modified action-dependent heuristic dynamic programming (ADHDP) framework. Unlike conventional adaptive assist-as-needed controllers based on Iterative Learning Control (ILC-AAN), the proposed RL-AAN controller autonomously adjusts the trade-off between movement errors and robot assistance in response to the user's recent performance, in real-time, while relying on a small set of high-level tunable parameters that do not require subject-specific manual adjustments. The RL-AAN controller was implemented on a cable-driven, end-effector type rehabilitation robot and validated against a conventional ILC-AAN controller through perturbation-based reaching tasks involving a group of healthy individuals. Compared to ILC-AAN, the RL-AAN controller significantly reduced the amount of robot assistance required during training, promoting user active participation and task performance. Following training with the RL-AAN controller, retention tests showed more accurate arm-reaching trajectories compared to ILC-AAN training, highlighting the potential of RL-AAN for future use in exercise-based rehabilitation. Overall, this work contributes to ongoing research into developing control strategies that enable personalization in physical human-robot interaction (pHRI) and robot-assisted rehabilitation.**

***Index Terms*— Assist-as-needed control, reinforcement learning control, rehabilitation robotics, physical human-robot interaction, cable-driven parallel robot.**

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I. Introduction

REHABILITATION of motor function is a fundamental component of the recovery process for individuals with movement impairments resulting from central or peripheral nervous system injuries (e.g., stroke, traumatic brain injury, spinal cord injury), neurodegenerative diseases, or musculoskeletal injuries [1]. Intense, repetitive, and task-oriented exercises can help individuals with brain injury regain lost motor functions through the process of motor (re)learning [2], [3]. However, traditional rehabilitative interventions are labor-intensive and physically demanding for therapists and face constraints due to limited availability and the high cost of trained personnel, which restrict the frequency and intensity of exercise sessions. Robot-assisted rehabilitation aims to retrain movement planning, control, and execution through exercises mediated by a robotic trainer that delivers controlled assistive or resistive forces, often in conjunction with real-time multisensory feedback, and records quantitative data to inform targeted therapy. This technology has the potential to reduce the physical burden on therapists, optimize human resource allocation, and standardize rehabilitation programs [4], while also empowering patients to engage in self-administered, home-based exercises [5].

Because active effort is a critical enabler of motor learning [6], it has been postulated that a robotic trainer should only provide the minimum level of assistance required for the patient to accomplish a desired motor task. Thus, an ideal device should be transparent (i.e., exert no force on the user) when the user performs well, and intervene only when the user encounters difficulty completing the task, in order to correct excessive movement errors. Control strategies that align with this principle are referred to as "assist-as-needed" (AAN) [7], "patient-in-charge" [8], or "patient-cooperative" [9] controllers. Pioneering AAN strategies relied on conventional impedance controllers (IC) with fixed stiffness and damping parameters, implementing a static relationship between movement errors and robot assistance [9], [10]. Such controllers require extensive manual tuning as the robot's compliance must be adjusted to the patient's varying motor skills over the therapy program [11]. To overcome this drawback, later

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works introduced adaptive AAN controllers that autonomously adjust the stiffness of an underlying IC based on the patient's most recent performance, leveraging a proportional iterative learning control (ILC) law [7], [12], [13]. Starting from a naive (untrained) state, an ILC-AAN controller updates the impedance parameter at each movement repetition by scaling the previous impedance parameter by a constant forgetting factor and adding a given function of the kinematic error measured at the previous repetition, multiplied by a constant gain. Other relevant examples of adaptive AAN controllers include: the error-based controller with an AAN force decay term introduced in [14], which modulates robot assistance by learning a model of the patient's impairment across the robot's workspace using radial basis functions; the joint-space controller developed in [15], which adjusts IC stiffness based on human net joint torques estimated from on-board load cells via a linear adaptation map; and the optimal controller described in [16], where IC stiffness is set inversely proportional to the estimated human joint stiffness, to minimize the weighted sum of kinematic errors and robot's effort. While all these approaches can gradually modulate the robot's assistance as the user's performances change, they share the following limitations: (i) they assume that the human motor system will always respond to robot-applied forces in the same stereotypical fashion, regardless of the therapy dosage already delivered–that is, the cumulative duration, intensity, and repetition frequency of prior training. This assumption fails to capture the non-linear, session-to-session learning dynamics observed during multi-session exercise, particularly in individuals with sensory or motor impairments; (ii) they are blind to previous movement executions, except for the most recent one; (iii) they operate with a fixed learning update rate, which must be manually adjusted to match the user's learning pace at different stages of the rehabilitation program.

To address the limitations of conventional adaptive AAN controllers, more recent studies have aimed to modulate the level of assistance provided by a robotic trainer *on-the-fly*, using more advanced learning-based control architectures. Sharifi et al. developed a time-varying impedance controller with iterative learning-based adaptation laws to adjust the robot impedance based on human behavior [17]. This method seeks to optimize the balance between movement accuracy and stability, facilitating safe physical human-robot interaction without requiring force/torque measurements. However, it does not directly encourage user participation. Obayashi et al. proposed a task-oriented AAN controller to help users learn a dart-throwing task [18]. The AAN controller adopts a policy-gradient reinforcement learning (RL) algorithm to adjust the IC stiffness based on the most recent throw score. However, this method does not modulate the relative importance of performance versus assistance as training progresses; thus both naive and experienced individuals are challenged using the same predefined control objective. Pareek et al. [19] proposed an RL–based AAN controller for handwriting rehabilitation, which was evaluated in a convenience sample of healthy individuals. This controller is trained offline to derive a subject-independent policy that is not updated at runtime. As a result, the policy cannot autonomously adapt to inter-individual variability or session-to-session performance changes without retraining or manual parameter adjustment. Our group introduced the first stride-dependent [20], [21] and phase-dependent [22] RL-AAN controllers for robot assisted gait training, which autonomously reshape the IC of an ankle exoskeleton in real-time based on the wearer's training performance. Subsequently, other researchers have proposed related RL-based strategies for lower-extremity exoskeletons [23], [24], [25].

Building upon the RL-AAN controller presented in [20] and [21], in this paper we leverage the vast expressive capability of actor-critic RL architectures to model the complex, subject-specific co-adaptation processes originating from the human-robot coupled system during repeated robot-assisted arm-reaching exercises. Although RL has been applied to functional electrical stimulation (FES) controllers for exercise-based rehabilitation and, more recently, to a variety of upper- and lower-limb assistive and rehabilitation robots—including powered orthoses, prostheses, and end-effector-based devices [26]—work on RL-based AAN control remains limited. To the best of our knowledge, this paper introduces the first online RL-based AAN control strategy for rehabilitation robots explicitly designed to promote *both* movement accuracy *and* user active effort during upper-limb reaching exercises. From an algorithmic standpoint, the new RL-AAN controller extends previous research by explicitly accounting for the user active effort in the actor's cost function, by modulating the balance between movement errors and robot assistance directly, through the actor's control objective, and by autonomously varying the task complexity in response to the subject's performance over recent movement repetitions following a performance-based curriculum learning scheme [27], [28], [29]. The resulting RL-AAN controller relies on a minimal set of subject-independent parameters, allowing for intuitive tuning and fast convergence. To demonstrate feasibility, we compared the effectiveness of the new RL-AAN controller to a conventional ILC-AAN controller, in terms of human motor adaptation during and immediately following sessions of robot-assisted arm-reaching tasks, which were administered using a custom-engineered cable-driven parallel robot (CDPR). The rest of this paper is organized as follows: Section II introduces the experimental setup, including the CDPR setup and the implementation of robot-assisted arm-reaching exercises; Section III describes the proposed RL-AAN controller and the conventional ILC-AAN controller (used for comparison) which modulate the CDPR's impedance during the arm-reaching exercises; Section IV deals with the experimental protocol used for validation with healthy individuals, and the statistical analysis. Results are presented in Section V and discussed in Section VI. Finally, conclusions and future work are presented in Section VII.

II. Methods

A well-established robot-assisted rehabilitation exercise for elbow and shoulder after brain injury consists of a sequence of planar point-to-point arm-reaching tasks along a prescribed set of directions [30]. Using the affected arm, patients are asked to repeatedly move the end-effector along reference straight-line paths towards targets displayed on a computer screen in a random sequence. Studies on motor skill learning

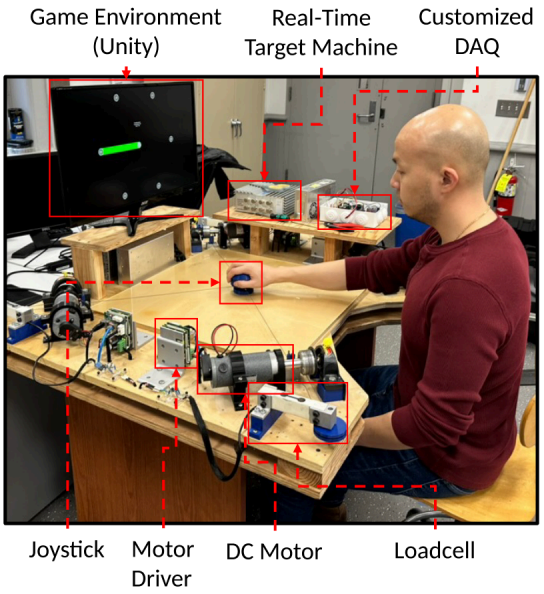


Fig. 1. Prototype of cable-driven parallel robot (CDPR) developed to evaluate the RL-AAN controller.

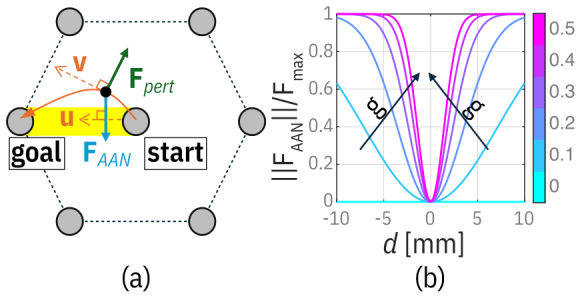


Fig. 2. (a) Schematic representation of the arm-reaching exercise. The objective is to move the joystick (black dot) from the initial position (start) to one of six possible target positions (goal) along a prescribed straight path (yellow highlight), whose direction is indicated by the unit vector **u**. During the movement, the user must compensate for a perturbation force ($\mathbf{F}_{pert}$) that is orthogonal to the instantaneous velocity of the joystick ($\mathbf{v}$) and proportional to its magnitude, as described by (1). An assistive force F_{AAN}, generated by a nonlinear impedance law (2) and orthogonal to u, helps the user overcome the perturbation. (b) Normalized magnitude of F_{AAN} as a function of the adjusted distance d and the impedance parameter g.

have adopted a similar training paradigm, adding mechanical perturbations to explore how healthy individuals adapt their movement patterns over hundreds of repetitions in response to disturbances [31]. To implement the rehabilitation task, we developed a planar CDPR inspired by a previous design described in [32]. Compared with common rigid-link designs, cable-driven devices show advantages in terms of inertial properties, force exertion capabilities, and affordability [33].

A. Hardware Setup

The custom-engineered CDPR, shown in Fig. 1, uses four coated steel cables arranged in parallel to exert desired planar forces on its end-effector (joystick). The joystick can be moved by the user over a planar workspace ($630 \times 630\,\text{mm}^2$), similar to a computer mouse. Each cable is controlled by a spool actuated by a brushed DC motor (Groschopp PM6015, 57 W, 220 mNm rated torque) and equipped with an incremental encoder (US Digital E5-1000, 4000 CPR). An idle pulley mounted at the tip of a cantilever beam loadcell measures the tensile force in each cable and routes the cable towards a pair of vertically mounted idle rollers. Each pair of idle rollers is located at one vertex of the square workspace, serving as the effective cable exit points for the CDPR. The distal end of each cable is connected to the same point on the vertical axis of the joystick. The top part of the joystick can rotate freely about the same vertical axis to minimize the transmission of spurious moments during motion. An acrylic plate covering the CDPR workspace and a low-friction adhesive film attached to the bottom surface of the joystick minimize the effect of sliding friction. A Real-Time Target Machine (SpeedGoat Baseline) operating at 1kHz serves as the CDPR controller and handles communication with the motor drivers (A-M-C DZEANTU-020B200) and a custom-engineered data acquisition (DAQ) board via an EtherCAT fieldbus. A 2D virtual playground developed in Unity provides the user interface for the arm-reaching exercises. Task-relevant data are sent from the CDPR controller to the Unity environment at 60 Hz through a User Datagram Protocol (UDP) socket.

B. Robot-Assisted Arm-Reaching Exercise

A serious game was implemented in the CDPR to challenge the user's ability to counter an external perturbation force during arm-reaching tasks. Starting from the center of the workspace, the user moves the joystick along a straight-line path displayed on a computer screen, as accurately as possible, while being subjected to the perturbation, to reach one of six predefined targets arranged at the vertices of a regular hexagon, located 20 cm from the starting position (Fig. 2-a). The user then brings the joystick back to the starting position and waits for the next target to appear at a random vertex. To pace the task, a blue indicator progresses from the center of the hexagon towards the current target within 1 second, following a 5th-order time law [34]. A 500 ms pause is inserted between the appearance of the next target on the screen, and the onset of the blue indicator motion. The color of the reference path is modulated by the current kinematic error d, serving as descriptive visual feedback. When the error is small ($d < 2.5\,\text{mm}$), the color is set to green. When the error is moderate ($2.5\,\text{mm} \leq d < 7.5\,\text{mm}$), the color progressively shifts from green to yellow, and when the error is high ($d \geq 7.5\,\text{mm}$), it gradually transitions to red. The external perturbation F_{pert} is defined as

$$F_{pert} = \gamma_{pert}\,\mathbf{v}_{\perp}, \tag{1}$$

where $\gamma_{pert} = 0.02\,\text{N/(mm/s)}$ is a scaling factor and $\mathbf{v}_{\perp}$ is the vector obtained by rotating the joystick's current velocity $\mathbf{v}$ by 90 deg clockwise (Fig. 2-a). Thus, the perturbation peaks when the upper limb moves the fastest. This perturbation strategy has been validated in prior studies as an effective way to introduce disturbance in the user's arm movement while ensuring safety and avoiding fatigue [35]. In addition to the perturbation, the CDPR exerts an assistive force defined as

$$F_{AAN} = -F_{\max}\left[1 - e^{-(gd)^2}\right]\mathbf{u}_{\perp}, \tag{2}$$

where $\mathbf{u}_{\perp}$ is obtained by rotating the unit vector $\mathbf{u}$ directed along the reference path and pointing towards the target by

TABLE I
ILC-AAN AND RL-AAN PARAMETERS

AAN Type	Parameter	Definition	Value
ILC	K	gain	0.015 mm^{-2}
	γ_{ILC}	forgetting factor	0.82
	b	intercept	-0.04 mm^{-1}
RL	α_a	Actor learning rate	0.5
	α_c	Critic learning rate	0.1
	γ_{RL}	Critic discount factor	0.5
	Λ	reward weight matrix	diag $([1, 1, 1.5, 1.5])$
	β	U_c step	0.01
	w_U	U_c window size	6 cycles
	η	acceptable error	$\{5, 3.5, 2\}$ mm^2
	w_η	η window size	$\{6, 12, 30\}$ cycles

90 deg clockwise, and $F_{\max} = 25$ N is the maximum assistive force dictated by the saturation of the CDPR's motors, whose magnitude is in line with past rehabilitation robots for the upper extremities [36]. Equation (2) creates a symmetric, nonlinear elastic virtual wall around the reference path, whose magnitude approximately equals F_{max} outside the wall and gradually decreases to a null value inside it (Fig. 2-b). The slope of the nonlinear wall is controlled by the impedance parameter g. The *adjusted* distance d is obtained from the *geometric* distance between the joystick's current position and the reference path, indicated as $\tilde{d}$, by considering a deadband $d_{\text{db}} = 3$ mm according to the following law:

$$d = \begin{cases} \text{sign}(\tilde{d})(|\tilde{d}| - d_{\text{db}}) & \text{if } |\tilde{d}| \geq d_{\text{db}} \\ 0 & \text{if } |\tilde{d}| < d_{\text{db}} \end{cases} \tag{3}$$

The deadband d_{db} was introduced to allow for human variability [7], and its value was selected based on the average standard deviation observed during preliminary tests with eight healthy individuals who performed the reaching tasks under null assistive or perturbation forces.

III. Assist-as-Needed Control

In this work, we investigate the modulation of the impedance parameter g in (2) by an adaptive AAN controller. We compared a novel RL-AAN controller against a traditional ILC-AAN controller and hypothesized that the enhanced versatility in adapting to user performances afforded by the learning-based method would result in improved human motor adaptation dynamics during the exercise and better retention after training with the assistive force. For both controllers the impedance parameter g was allowed to vary within the interval $[0, 0.5]$ mm^{-1} (Fig. 2-b).

A. ILC-AAN Controller

ILC is often utilized in robotic applications involving repetitive tasks. ILC aims to improve task execution by gradually adjusting the control action after each task repetition, leveraging feedback from previous repetitions [12]. In the present implementation, a P-type ILC-AAN controller [7], [37] adjusts the impedance parameter g to be applied during the next movement repetition (hereafter referred to as "cycle") according to

$$g_{k+1} = \gamma_{ILC}\, g_k + f(d_k), \tag{4a}$$
$$f(d_k) = K\, d_k - b, \tag{4b}$$

where k is the index of the most recently completed cycle, $f(d)$ is a linear function of the adjusted movement error d, with gain $K > 0$ and intercept b, and $\gamma_{ILC} \in (0, 1)$ is the forgetting factor. Appropriate tuning of γ_{ILC}, K, and b is critical to ensure adaptability to user performance and promote active participation. The forgetting term γ_{ILC} must be adjusted to prevent the user from over-relying on the CDPR assistance, thereby reducing their own effort [38]. Indeed, γ_{ILC} determines the rate at which the learning algorithm reduces g when the movement error is small, steering the CDPR towards a weaker assistive force field (2), with the effect of challenging the user. This fading assistance has positive effects in terms of motor learning [39]. If K is not properly tuned in relation to γ_{ILC}, the controller may not be able to compensate for the effect of the forgetting factor. This could decrease g even when the error is substantial, or, conversely, saturate g even at small errors, effectively leading to a static IC. In this work, we selected the ILC parameters as shown in Tab. I, based on preliminary tests.

Despite its popularity, ILC-AAN requires extensive trial-and-error tuning, and its myopic control law (4), which accounts only for the user's most recent performance, may fail to effectively replicate the complex dynamics of human motor adaptation. This motivates the development of a new learning-based AAN controller, as described in Sec. III-B.

B. RL-AAN Controller

Unlike other optimal control strategies, RL controllers learn an optimal policy *on-the-fly*, while interacting with the environment [40]. This is especially convenient when an accurate model of the environment is difficult to obtain, as is the case of physical human-robot interaction (pHRI) involving complex human-robot dynamics. Herein, we relied on the actor-critic RL framework [41], whereby the actor determines the most suitable action at each cycle k, based on the system states and feedback from the critic, while the critic assesses the effectiveness of that action in terms of the anticipated long-term costs. In the context of RL control, the long-term cost at the k-th cycle, denoted as $V(k)$, is defined as the discounted sum of all future negative rewards [42]

$$V(k) = \sum_{i=0}^{\infty} \gamma_{RL}^{i}\, r(k + i + 1), \tag{5}$$

where $r(k)$ represents the immediate negative reward incurred at the k-th cycle, and $\gamma_{RL} \in [0, 1]$ is the discount factor. The basic idea behind Heuristic Dynamic Programming (HDP) is to estimate $V(k)$ via function approximation [43]. Following this approach, the critic neural network (NN) estimates the approximated long-term cost $\tilde{V}(k)$ at the k-th cycle by minimizing the temporal difference error $t_d(k)$ derived from the Bellman equation [44]:

$$t_d(k) = r(k) + \gamma_{RL}\tilde{V}(k) - \tilde{V}(k-1) \tag{6}$$

In this work, we adopted the Action-Dependant HDP (ADHDP) [40], a modified HDP algorithm that is well suited for complex systems and features convenient uniformly ultimate boundedness (UUB) properties [45]. Following the ADHDP method, the critic NN estimates $\tilde{V}(k)$ as a state-action-value function, considering both the states and the

action as inputs (Fig. 3). Thus, the optimal policy is the one minimizing $\tilde{V}(k)$. In the present implementation, the RL-AAN model is characterized by:

- States $S(k) = [\bar{e}_{pos}(k), \bar{e}_{neg}(k), e^{+}_{pos}(k), e^{+}_{neg}(k)]$, where $\bar{e}_{pos}(k)$, $e^{+}_{pos}(k)$ and $\bar{e}_{neg}(k)$, $e^{+}_{neg}(k)$ are the mean (indicated by the overline) and peak (indicated by the + superscript) positive and negative errors measured during the k-th cycle. Positive errors are those in the same direction of the perturbation force, F_{pert}. Negative errors are those opposite to F_{pert}. Thus, positive and negative errors indicate under-correction and over-correction by the user, respectively.
- A set of possible actions $g(k) \in [0, 0.5]$ mm^{-1}
- An immediate cost $r(k) = S^T(k)\,\Lambda\,S(k)$, where $\Lambda \in \mathbb{R}^{4\times4}$ is a positive-definite diagonal weight matrix.

Both the actor and the critic networks are nonlinear feed-forward NNs with one hidden layer and sigmoid activation functions [40]. The critic NN is parameterized as

$$\tilde{V}(k) = W_c^{(2)}(k)\sigma(H_c(k)), \tag{7a}$$

$$H_c(k) = W_c^{(1)}(k)\left[S^T(k), \delta_a(k)\right]^T, \tag{7b}$$

where $W_c^{(1)}(k)$ and $W_c^{(2)}(k)$ are the weights matrices for the hidden and output layers, respectively, $H_c(k)$ is the hidden layer that connects $W_c^{(1)}(k)$ and $W_c^{(2)}(k)$, and $\delta_a(k)$ is the output of the actor NN. The gradient descent method updates the critic NN weights, with a learning rate α_c, to minimize the temporal difference error (6) according to the quadratic cost $E_c(k)$:

$$E_c(k) = \frac{1}{2}t_d^2(k) \tag{8}$$

The weights of the actor NN are updated using the gradient descent method with learning rate α_a to minimize the actor cost function $E_a(k)$, defined as

$$E_a(k) = \frac{1}{2}\left[\tilde{V}(k)\,(1 - U_c(k)) + g(k)\,U_c(k)\right]^2, \tag{9}$$

where $U_c(k)$ is a specified control objective. Because the actor output $\delta_a(k)$ is bounded to the interval $[0, 1]$ by the sigmoid activation function, a scaling factor $\rho = 0.5$ mm^{-1} is introduced to map the impedance parameter to the feasible interval $[0, 0.5]$ mm^{-1}:

$$g(k) = \rho\,\delta_a(k) \tag{10}$$

The control objective $U_c(k)$ in (9) is updated after each cycle based on the user's performance over the last w_U cycles, through the following law

$$U_c(k+1) = \begin{cases} \min(1,\ U_c(k) + \beta) & if\ \bar{r}(k) \le \eta(k) \\ \max(0,\ U_c(k) - \beta) & if\ \bar{r}(k) > \eta(k), \end{cases} \tag{11}$$

where $\bar{r}(k)$ is the arithmetic mean of the immediate costs measured over the last w_U cycles, β is the rate of change of $U_c(k)$, and $\eta(k)$ indicates the current level of *acceptable error*. We note that $U_c(k)$ is constrained to the interval $[0, 1]$ by definition. Thus, the actor cost (9) is the squared convex combination of $\tilde{V}(k)$, representing combined kinematic errors, and $g(k)$, indicating the current level of robot assistance, with their relative importance modulated by $U_c(k)$. In essence, U_c controls the balance between two conflicting objectives in exercise training: minimizing external assistance to encourage active effort, vs. minimizing movement errors to promote correct task execution. When the user performs well, the controller increases U_c, thereby accepting larger kinematic errors and further penalizing robot guidance, which results in a more compliant joystick. Conversely, when movement errors increase, the controller decreases U_c, stepping up the assistive action to reduce kinematic errors.

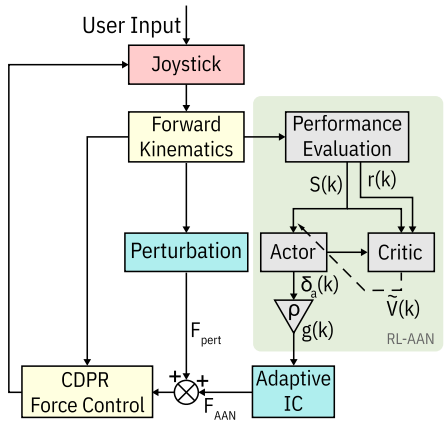


Fig. 3. Structure of the RL-AAN controller. $S(k)$: state vector at cycle k; $r(k)$: immediate cost; $\tilde{V}(k)$: approximated long-term cost; $\delta_a(k)$: actor output; ρ actor scaling factor; $g(k)$: impedance parameter; IC: impedance controller; CDPR: cable-driven parallel robot.

As can be inferred from (11), a sustained value of $U_c(k) = 1$ indicates consistently acceptable performance, whereas a consistently low $U_c(k) < 1$ in recent executions suggests that the task has become too challenging for the user. Leveraging this insight, we update the level of acceptable error $\eta(k)$ based on the user performance measured over the last w_η cycles, according to the following law:

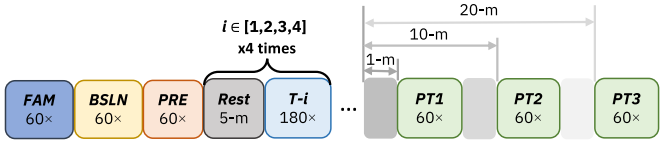
$$\eta(k+1) = \begin{cases} \max(\eta_{\text{low}},\ \eta(k) - \zeta)\ \text{if}\ U_c(k-i) = 1\ \forall\, i \in [0, w_\eta] \\ \min(\eta_{\text{high}},\ \eta(k) + \zeta)\ \text{if}\ \nexists\, i \in [0, w_\eta]\ \text{s.t.}\ U_c(k-i) = 1 \\ \eta(k) \qquad\qquad\qquad\qquad \text{otherwise} \end{cases} \tag{12}$$

As indicated in (12), if all U_c values within the assessing window w_η are unitary, a more challenging objective is set by decreasing η by a fixed quantity ζ. However, if the user never reaches a unitary U_c in the last w_η cycles, η is increased by the same quantity to reduce task difficulty. Based on preliminary tests, we determined that $\eta_{low} = 5$ mm and $\eta_{high} = 2$ mm were perceived as low and high-difficulty levels, respectively. Thus, by setting $\zeta = 1.5$ mm, the controller can switch across three equally-spaced levels of difficulty: η_{low}, η_{med}, η_{high}. Additionally w_η was set to increase as η drops ($w_{\eta,low} = 6$, $w_{\eta,med} = 12$, $w_{\eta,high} = 30$) to encourage users to explore more challenging motor tasks. The architecture of the proposed RL-AAN controller and its parameters are reported in Fig. 3 and Tab. I, respectively.

IV. Experimental Validation

A. Protocol

Fifteen healthy adults (11 males, age 18-32 years) volunteered for this study. Participants had no musculoskeletal



	FAM	BSLN	PRE	T-i	PT-i
F_{AAN}	✗	✗	✗	✓	✗
F_{pert}	✗/✓	✗	✓	✓	✓

Fig. 4. Experimental Protocol (FAM: familiarization; BSLN: baseline; PRE: pre-training; T-i: i-th training; PT1-PT3: post-training). Symbols ✓ and ✗ indicate whether the assistive (F_{AAN}) or perturbation (F_{pert}) forces generated by the CDPR were enabled/disabled.

or neurological injuries that would affect their ability to complete reaching tasks with their dominant arm. The study was approved by the Institutional Review Board at Stevens Institute of Technology (Protocol 2022-034), and all participants provided written consent prior to data collection. The study included two visits, one for each controller (ILC-AAN and RL-AAN), scheduled at least 48 hours apart to ensure a sufficient washout period [46]. To account for any potential residual carry-over effects not fully eliminated by the washout period, the order of the controllers was balanced across participants using a Latin square design.

The protocol followed in each visit is illustrated in Fig. 4. First, each participant underwent a 60-cycle familiarization session (FAM), consisting of 30 arm-reaching repetitions without F_{pert}, and 30 repetitions with F_{pert}, spanning all six directions, in random order. This session was carried out with null F_{AAN} to let the participant familiarize themselves with the task and the type of perturbation generated by the CDPR. Next, participants completed a baseline session (BSLN) consisting of 60 arm-reaching cycles (10 along each direction, in random sequence) performed with null F_{AAN} and F_{pert}, to evaluate their unperturbed/unassisted task performance. Subsequently, participants underwent a 60-cycle pre-training (PRE) session, during which they were exposed to F_{pert} without receiving external assistance. The purpose of this session was to evaluate how each participant responded to unexpected disturbances. Data collected from this session was also compared to post-training data to evaluate any training effects. Additionally, for the RL-AAN controller, the PRE session was used to pre-train the critic NN, starting from a naive state and setting $g \equiv 0$ in place of the actor output. This approach improves data efficiency by allowing the critic NN to observe the environment under a transparent policy [19]. Afterwards, participants took part in 4 training sessions (T-1 to T-4), each consisting of 180 cycles (i.e., 30 cycles along each direction, in random order) with both F_{pert} and F_{AAN} active. The RL-AAN and ILC-AAN controllers adapted cycle-by-cycle throughout the entire training period (T-1 to T-4) without being reset. Five-minute breaks were included in-between PRE and T-1 to T-4 sessions to minimize physical and mental fatigue. To assess the development of internal feed-forward models in the study participants, 12 catch trials – evenly distributed across the 6 movement directions – were administered during each training session. Specifically, one catch trial was randomly inserted into each 30-cycle block by unexpectedly removing both F_{AAN} and F_{pert}, as described in [47]. These 12 catch trials were conducted in addition to the 180 cycles that comprised each training session. Finally, training effects were assessed during three post-training sessions (PT1 to PT3), each including 60 cycles with F_{pert} but no assistance. These sessions began 1, 10, and 20 minutes after the end of T-4 (Fig. 4), respectively, in line with previous studies [22], [48].

B. Statistical Analysis

To compare the effectiveness of the two controllers in promoting motor adaptations during training, a linear mixed model (LMM) analysis was carried out with *Controller* (ILC-AAN, RL-AAN) and *Session* (T-1 to T-4) as independent variables. *Intercept*, *Session*, *Controller*, and Session × Controller were set as fixed effects, and *Intercept* was also set as a random effect, to account for individual differences in initial performance:

$$Y_{ij} = \beta_0 + \beta_1 S_{ij} + \beta_2 C_{ij} + \beta_3 S_{ij} C_{ij} + u_{0i} + \varepsilon_{ij} \tag{13}$$

In the above equation, i and j are the indices of the study participant and training session, respectively; $S_{ij} \in \{0, 1, 2, 3\}$, and $C_{ij} \in \{0, 1\}$ encode *Session* and *Controller*, respectively; u_{0i} is the subject's random *Intercept*, and $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ represents residuals [49]. The dependent variables included the average impedance parameter g, evaluated over each T session and used as a surrogate measure for robot assistance, and the percent changes from PRE to each T session in the mean positive and negative errors ($\Delta\bar{e}_{pos}\%$, $\Delta\bar{e}_{neg}\%$), which quantified changes in an individual's tendency toward under-correction and over-correction. To assess whether there was a decline in η over the training sessions, indicating performance improvement when using the RL-AAN controller, we conducted an LMM analysis on the total number of cycles executed at each η. The model included *Movement Cycle* as a fixed effect, and *Intercept* as both a fixed and random effect:

$$Y_{ij} = \beta_0 + \beta_1 M_{ij} + u_{0i} + \varepsilon_{ij} \tag{14}$$

In the above equation, M_{ij} encodes the *Movement Cycle*, while all other parameters are defined as in (13). As a secondary analysis, catch trials were examined using one sample t-tests applied to the differences in mean negative errors between each catch trial and the cycle immediately preceding it.

Next, to establish the presence of learning effects, we carried out a third LMM analysis, similar to (13), using *Intercept*, *Session* (PT1 to PT3), *Controller*, and *Session* × *Controller* as fixed effects, *Intercept* as a random effect, and percent differences in the error metrics between PRE and each PT session ($\Delta\bar{e}_{pos}\%$, $\Delta\bar{e}_{neg}\%$) as dependent variables. This analysis evaluated subjects' short-term retention, defined as their ability to track the target motion while compensating for the perturbation in the absence of robotic assistance.

The assumption of normality was assessed by inspecting QQ-plots of the residuals (for LMM analyses) and by applying the Shapiro-Wilk test (for one-sample t-tests). Potential correlations between $\Delta\bar{e}_{pos}\%$ and $\Delta\bar{e}_{neg}\%$, were assessed

TABLE II
LMM RESULTS FOR TRAINING AND RETENTION

		INTERCEPT ILC-AAN		INTERCEPT RL-AAN		SESSION ILC-AAN		SESSION RL-AAN		CONTROLLER RL-AAN – ILC-AAN		SESSION×CONTROLLER RL-AAN – ILC-AAN	
		β (SE)	p	β (SE)	p	β (SE)	p	β (SE)	p	β (SE)	p	β (SE)	p
Training	$\Delta\bar{e}_{pos}$%	**53.9 (11.9)**	**<.001**	**55.1 (11.9)**	**<.001**	–	ns	**4.4 (1.8)**	**.019**	–	ns	–	ns
	$\Delta\bar{e}_{neg}$%	–	ns	–	ns	–	ns	**-12.9 (6.0)**	**.034**	–	ns	–	ns
	g	**0.10 (0.02)**	**<.001**	**0.06 (0.02)**	**.003**	–	ns	–	ns	**-0.04 (0.008)**	**<.001**	–	ns
Retention	$\Delta\bar{e}_{pos}$%	**35.8 (14.3)**	**.014**	**64.1 (14.3)**	**<.001**	–	ns	–	ns	**28.3 (6.4)**	**<.001**	–	ns
	$\Delta\bar{e}_{neg}$%	–	ns	–	ns	–	ns	–	ns	–	ns	–	ns

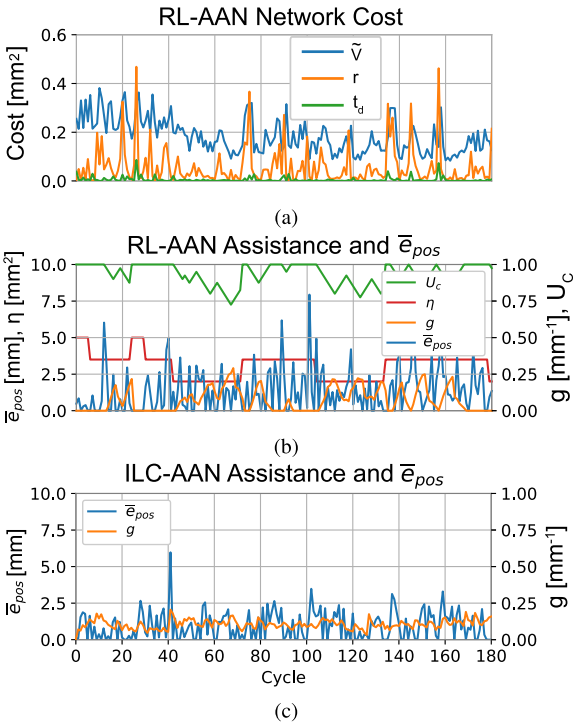


Fig. 5. (a) Adaptation of the critic NN in the RL-AAN controller during T-1 for a representative study participant; (b) Corresponding trajectories of U_c, η, and g, and resulting kinematic errors $\bar{e}_{pos}$; (c) Adaptation of g and resulting kinematic errors $\bar{e}_{pos}$ for the ILC-AAN controller during T-1 for the same participant.

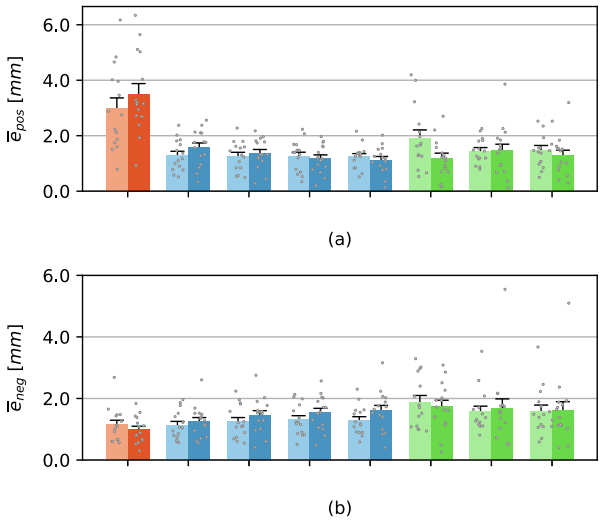


Fig. 6. Group averages of the error metrics $\bar{e}_{pos}$ and $\bar{e}_{neg}$ during the training protocol for the ILC-AAN controller (lighter shade) and the RL-AAN controller (darker shade). Overlaid scatter points indicate individual participant values. Error bars: +1 SE (standard error).

TABLE III
LMM FOR η

	β (SE)	p
Intercept	3.053(.018)	$< .001$
Cycle	$-3.35 \times 10^{-4}(4.93 \times 10^{-5})$	$< .001$

using Spearman's rank correlation coefficients (ρ). A diagonal covariance structure was selected for the LMM analyses to account for potential deviations from homoskedasticity. All samples (cycles), except those from the catch trials, were used to compute the subject- and session-specific averages for the LMM analyses. For the catch-trial analysis, each subject's responses across the six directions were pooled to obtain a subject-specific mean value, which was then used in the one-sample tests. The significance level was set to $\alpha = .05$, and all data analyses were performed using IBM SPSS Statistics 29.0.

V. RESULTS

A. Motor Adaptation During Training

Figure 5a shows the trajectories of the immediate cost r, the temporal difference error t_d, and the approximated long-term cost $\tilde{V}$, measured from a representative participant during T-1. It can be observed that t_d is consistently small across movement repetitions (cycles), indicating an accurate estimation of $\tilde{V}$. As can be inferred from the cycle-by-cycle $\bar{e}_{pos}$ in Fig. 5b, the trend in U_c depends on recent kinematic errors. When U_c is unitary, no assistance is provided by the RL-AAN controller (i.e., null g). Conversely, when U_c decreases, the actor NN responds by increasing g. Additionally, sustained good performance causes the RL-AAN controller to increase the task difficulty (i.e., decrease η), while degrading performance has the opposite effect. Figure 5c shows the trajectories of g and $\bar{e}_{pos}$ obtained with the ILC-AAN controller, during T-1, for the same representative subject. Compared with the RL-AAN controller, the ILC-AAN controller produced generally larger and more stable g (ILC-AAN: 0.10 ± 0.03 mm^{-1}, RL-AAN: 0.07 ± 0.08 mm^{-1}), indicating stiffer and less responsive assistance.

At the group level, QQ-plots of the residuals and Shapiro-Wilk tests did not reveal significant deviations from normality. Spearman's rank correlation coefficients indicated no significant association between $\Delta\bar{e}_{pos}$% and $\Delta\bar{e}_{neg}$%, supporting the assumption that under-correction and over-correction represent distinct motor constructs. Figure 6a shows the group averages of $\bar{e}_{pos}$ for the ILC-AAN and RL-AAN controllers across all sessions. Both controllers induced significant reductions in $\bar{e}_{pos}$ between PRE and T-1, as confirmed by the positive intercepts of $\Delta\bar{e}_{pos}$% ($p < .001$), which did not differ

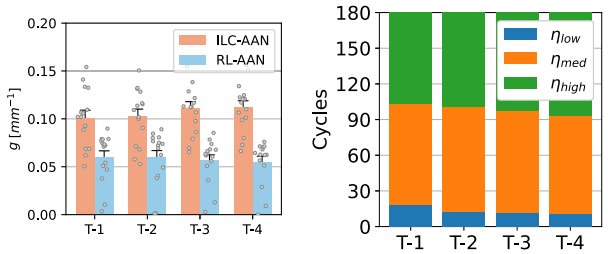


Fig. 7. (a) Group averages of the mean impedance parameter g across training sessions. Overlaid scatter points indicate individual participant values. (b) Occurrence of η across the RL-AAN training sessions. Error Bars: +1 SE.

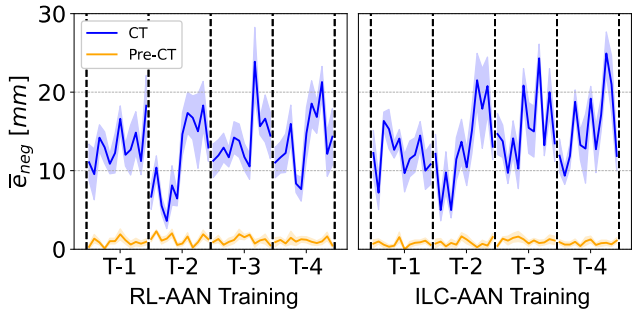


Fig. 8. Group averages of $\bar{e}_{neg}$ at the cycle immediately preceding (yellow, Pre-CT) and during (blue, CT) each of the 12 catch trials randomly inserted in each training session, for the two AAN controllers. Shaded areas indicate ±1 SD (standard deviation).

substantially between the two controllers (Tab. II). Yet, the RL-AAN controller demonstrated a more compliant behavior than the ILC-AAN controller ($p < .001$), as indicated by the results on g shown in Tab. II and Fig. 7a. Interestingly, the LMM analysis also revealed a significant improvement in $\bar{e}_{pos}\%$ from T-1 to T-4 for the RL-AAN controller ($p < .05$), which was accompanied by a significant increase in $\bar{e}_{neg}\%$ ($p < .05$). Together, these trends indicate a clear shift from undercompensation to overcompensation that was not observed in the ILC-AAN controller (Tab. II). Such difference suggests that the RL-AAN controller was more effective in promoting exploratory learning and active engagement.[1] Additionally, Fig. 7b suggests that the number of task executions performed at high task difficulty (η_{high}) increased, while the number of executions at low task difficulty (η_{low}) decreased as the RL-AAN training progressed. This was confirmed by the significant decrease in η from T-1 to T-4 ($p < .001$) revealed by the LMM analysis in Tab. III. Given that the value of η is linked to task performance according to (11) and (12), these trends emphasize the RL-AAN controller's capacity to induce motor adaptation during training.

Catch trials were conducted on a subset of study participants (N=6). One-sample t-tests (Tab. IV) revealed significant increases in $\bar{e}_{neg}$ ($p < .001$) between the cycle immediately preceding each catch trial and the cycle of the catch trial for both controllers. These marked kinematic deviations in the direction opposite to the expected perturbation provide evidence of internal feed-forward model formation associated with human motor adaptation. For reference, Fig. 8 shows group averages of $\bar{e}_{neg}$ for the cycles immediately before and during the catch trials.

[1]Herein, we use the expression "user engagement" to refer to the degree to which individuals actively participate in—and remain physically involved with—the motor task. A non-zero $\bar{e}_{neg}$ indicates that the participant exerted sufficient force to counteract or overcome the perturbation. Thus, $\bar{e}_{neg}$ was utilized as a proxy for engagement.

TABLE IV
ONE-SAMPLE T-TESTS FORCATCH TRIALS

	Controller	p-value	Mean, 95% CI
$\bar{e}_{neg}(k_c) - \bar{e}_{neg}(k_c - 1)$	ILC-AAN	$< .001$	13.08 [11.69, 14.48]
	RL-AAN	$< .001$	10.81 [8.82, 12.80]

k_c: Index of catch trial cycle.

B. Retention

The LMM analysis of the post-training data revealed a significant reduction in undercompensation (i.e., a positive $\Delta\bar{e}_{pos}\%$ intercept) relative to PRE performance immediately following training with RL-AAN ($p < .001$), which was retained across all three PT sessions without an appreciable decline in performance (Tab. II). Although a significant decrease in undercompensation was also observed immediately after ILC-AAN training ($p < .05$), this improvement was less pronounced than that observed after RL-AAN training ($p < .001$), indicating more accurate task execution following RL-AAN traininig. This was likely due to participants over-relying on the ILC assistance during training. Indeed, PT1 started just 1 minute after the end of T-4. As the stiff ILC assistance was suddenly removed, subjects had to re-adapt to effectively counteract F_{pert} without robot guidance. The RL-AAN method generated a more compliant assistance throughout the training (Fig. 7a), thus mitigating this undesirable after effect. In summary, the higher exposure to the perturbation during training with the RL-AAN controller elicited more accurate retention of the target movement following training, compared to the ILC-AAN controller, the gains in performance being largely driven by a lower tendency to undercompensate F_{pert}.

VI. DISCUSSION

In recent years, there has been increasing attention on rehabilitation technologies that enable the individualization of therapy procedures [50]. Traditionally, generalized control approaches have been deployed with limited success [38]; however, acknowledging the variability in individuals' behavioral profiles has unveiled new possibilities for robot-assisted treatment [51], [52], [53]. Robotic technologies are well-positioned to continuously evaluate an individual's motor performance and adjust exercise challenge levels accordingly, thereby promoting sustained engagement [54]. Integrating personalization in robot-assisted exercise training may facilitate the development of internal models, enhancing the effectiveness of rehabilitation therapies [55], [56].

This study introduced a new RL-based AAN controller for robot-assisted arm-reaching exercises, demonstrating that the new RL-AAN controller outperforms a conventional ILC-based adaptive controller in achieving the primary objective of AAN control: to self-adapt the level of assistance provided by a rehabilitation robot to an individual's performance, offering only the minimal assistance needed for them to complete

the motor task, to encourage active participation and promote motor adaptation. The conventional ILC-AAN controller relies on a simplistic law that only considers the most recent movement execution and always responds in the same standardized way to the user's motor behaviors, irrespective of the amount of training received. Conversely, the proposed RL-AAN controller encourages active participation at three different levels. First, unlike ILC-AAN, RL-AAN considers multiple error components in its states. By distinguishing overcompensation ($\bar{e}_{neg}$) from undercompensation ($\bar{e}_{pos}$), it can track user engagement after each movement execution and more effectively identify and exploit hidden patterns relating user participation to long-term performance. Second, the actor NN continuously seeks an optimal policy minimizing a varying cost that embeds both measures of an individual's engagement (physical effort) and estimates of their future motor behaviors (kinematic errors), leveraging the user's performance history to modulate the relative importance between the two contributions via the control objective U_c. Finally, the RL model features a performance-based curriculum learning structure [27], [28], [29], which adjusts the level of task difficulty by dynamically modifying the level of acceptable error η [21], thereby allowing the RL-AAN controller to not only adjust the assistance level but also to scale task difficulty based on user performance. This curriculum-learning-based adaptation mechanism promotes sustained engagement and may facilitate motor learning by keeping the task within the user's zone of proximal development, i.e., the space between what a learner can do without assistance and what they can do with assistance [57]. Overall, the LMM analysis indicated that the RL-AAN controller induced an increasing trend in user engagement during training, resulting in better retention after training compared to the conventional ILC-AAN controller.

Two studies on RL-based controllers for robot-assisted upper-limb rehabilitation incorporated minimization of robotic assistance into their model design [18], [19]. While both studies reported improvements in error reduction, they did not include a quantitative analysis of user engagement during training and retention sessions, nor did they administer catch trials to validate the establishment of internal models. In this context, the findings of the present study provide new insights, supporting our initial hypothesis that the enhanced adaptability of the RL-AAN controller may lead to more efficient learning compared to the conventional ILC-AAN method, as evidenced by improved human motor adaptation dynamics during the exercise and more accurate retention following training.

Due to its limited adaptability, the ILC-AAN controller is intrinsically difficult to fine-tune to an individual. The optimal forgetting factor and error gain can vary significantly across subjects, and must be determined by trial-and-error. This is an extremely tedious task, as it requires multiple rounds of testing and validation, with sufficient washout periods in-between trials. For this reason, most studies on the topic have adopted one-size-fits-all settings [7], [13]. In contrast, the proposed RL-AAN controller achieves personalization, without requiring individualized tuning, by incorporating the control objective U_c and the high-level performance parameter η. This approach accounts for user engagement and simplifies the model's tuning process. This architecture provides a leveled structure ensuring that both the actor and critic NNs converge rapidly, significantly enhancing the model's real-time adaptability to individual behaviors, and eliminating the need for manual tuning. Furthermore, U_c and η facilitate more intuitive performance monitoring and system adjustments.

This simplified tuning process offers an advantage not only over traditional ILC-based controllers, but also over more recent learning-based models. While RL-based algorithms generally require less parameters to be manually tuned compared to traditional ILC-based controllers, they also necessitate large pre-training datasets [18], [58], making them less practical for real-world applications. For example, the AR3n controller proposed in [19] requires a considerable amount of pre-training data and a simulated training environment, which adds another layer of complexity. In contrast, the proposed RL-AAN controller takes advantage of a curriculum-learning-based design that does not require any large pre-trained dataset. The critic NN converges during the PRE session, significantly lowering the demand of data and mitigating participant burden. In summary, through continuous re-balancing of kinematic error and robotic effort via the actor cost U_c, the proposed RL-AAN controller achieves user adaptation beyond what is afforded by traditional ILC laws and offline RL methods.

The present study also highlighted several limitations of the proposed RL-AAN framework. First, the acceptable error thresholds (η_{low}, η_{med}, and η_{high}) were manually set to uniform values across all participants, and the network learning rates (α_c, α_a) were defined from pilot data. This approach introduces subjectivity and may not generalize well to users with heterogeneous functional impairments. Developing automated methods to tune these thresholds and adapt learning rates based on individual performance would enhance both objectivity and generalizability. Second, future work should include a sensitivity analysis to examine the relative impact of the RL-AAN parameters in Tab. I in terms of user performance. In terms of experimental design, incorporating a control condition with visual feedback but no physical guidance would be valuable for elucidating how RL- and ILC-based assistance compare to unassisted practice.

While this study suggests the effectiveness of the RL-AAN algorithm in inducing short-term motor adaptations in healthy individuals, additional work is required before this approach can be piloted in individuals post-stroke. Stroke survivors often exhibit hemiparetic weakness, spasticity, abnormal muscle synergies, reduced shoulder and elbow range of motion, and sensory deficits —factors that can profoundly influence reaching task performance [59]. Notably, substantial direction-dependent differences in transverse-plane reaching have been reported in this population. Diagonal movements toward the contralateral-front quadrant and back toward the ipsilateral-back quadrant appear the most sensitive to impairment level [60], likely due to common deficits in elbow-shoulder coordination (e.g., flexion synergy). By contrast, findings for anteroposterior and mediolateral reaching tasks have been mixed [61], [62]. Therefore, future work involving stroke survivors should incorporate larger samples and increased repetitions per target to examine whether movement direction moderates performance during and after RL-AAN training.

Two recent meta-analyses of randomized clinical trials on robot-assisted upper-limb therapy post-stroke found significant immediate benefits of robotic therapy over standard care when total training time exceeded 15 hours, the robot operated in a "patient-active" mode (i.e., patient-cooperative, AAN, IC, or EMG-based control), and therapy was delivered to first-time stroke patients with mild to moderate impairment [63], [64]. However, both studies found no evidence of superiority of robotic over conventional therapy in follow-up assessments. Therefore, a future pilot cross-sectional study evaluating safety and proof-of-concept clinical feasibility of the RL-AAN controller should be followed by a longitudinal trial incorporating extended ($\geq$ 15 hrs.) robotic training and follow-up evaluations to determine whether the benefits of adaptive and personalized AAN robotic therapy are sustained over time. Furthermore, since rehabilitation gains are largely task-specific [2], further research is warranted to explore the application of the presented RL-AAN strategy to robotic platforms that enable reaching tasks beyond the transverse-plane motions investigated in this paper, as well as those enabling training of the distal upper limb [64].

VII. Conclusion

This work introduced the first on-line RL-based assist-as-needed strategy explicitly designed to promote both movement accuracy and user active effort during upper-limb reaching exercises. The proposed RL-AAN controller employs an ADHDP framework to modulate the level of robotic assistance based on the user's recent performance. Unlike other RL-based controllers for upper-limb rehabilitation, the proposed controller dynamically adjusts the balance between user performance and effort in the actor cost, as well as the task difficulty, to encourage active participation and promote motor adaptation.

The RL-AAN controller was validated against a conventional ILC-AAN controller through tests with healthy individuals using a cable-driven rehabilitation robot. During training, the RL-AAN controller outperformed the ILC-AAN controller by eliciting similar levels of movement accuracy on the user while applying substantially lower assistance. Furthermore, study participants demonstrated more accurate retention of the target movement after training with the RL-AAN controller.

This work contributes to ongoing research demonstrating the benefits of personalization in pHRI and robot-assisted training, paving the way for future validation studies with brain-injured populations.

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